

Important: Text generated by AI and may contain incorrect information, always check the content.

```
```python
Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

Display information about the dataframe
display('DataFrame shape:')
display(df.shape)
display('DataFrame columns and types:')
display(df.dtypes)
display('Missing values:')
display(df.isnull().sum())

Summary statistics for numerical columns
display('Summary statistics for numerical columns:')
display(df.describe())

Check the distribution of the target variable 'Diabetes_012'
display('Distribution of Diabetes_012:')
display(df['Diabetes_012'].value_counts())

Plot histograms for numerical columns
numerical_columns = df.columns
fig, axes = plt.subplots(nrows=len(numerical_columns), ncols=1, figsize=(10, 3*len(numerical_columns)))
for i, col in enumerate(numerical_columns):
 sns.histplot(df[col], ax=axes[i], kde=True)
 axes[i].set_title(col)
plt.tight_layout()
display('Histograms for numerical columns:')
plt.show()

Plot correlation matrix
display('Correlation matrix:')
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=False, cmap='coolwarm', square=True)
plt.show()

Check for highly correlated features
corr_matrix = df.corr().abs()
upper_tri = corr_matrix.where(np.triu(np.ones(corr_matrix.shape),k=1).astype(np.bool))
to_drop = [column for column in upper_tri.columns if any(upper_tri[column] > 0.95)]
display('Highly correlated features:')
display(to_drop)

Box plots to check for outliers
fig, axes = plt.subplots(nrows=len(numerical_columns), ncols=1, figsize=(10, 3*len(numerical_columns)))
for i, col in enumerate(numerical_columns):
 sns.boxplot(df[col], ax=axes[i])
 axes[i].set_title(col)
plt.tight_layout()
display('Box plots to check for outliers:')
plt.show()
```
```

'DataFrame shape:'

(253680, 22)

'DataFrame columns and types:'

0

| | |
|--------------|---------|
| Diabetes_012 | float64 |
|--------------|---------|



| 0 | |
|----------------------|---------|
| HighBP | float64 |
| HighChol | float64 |
| CholCheck | float64 |
| BMI | float64 |
| Smoker | float64 |
| Stroke | float64 |
| HeartDiseaseorAttack | float64 |
| PhysActivity | float64 |
| Fruits | float64 |
| Veggies | float64 |
| HvyAlcoholConsump | float64 |
| AnyHealthcare | float64 |
| NoDocbcCost | float64 |
| GenHlth | float64 |
| MentHlth | float64 |
| PhysHlth | float64 |
| DiffWalk | float64 |
| Sex | float64 |
| Age | float64 |
| Education | float64 |
| Income | float64 |

dtype: object
'Missing values:'

| 0 | |
|--------------|---|
| Diabetes_012 | 0 |
| HighBP | 0 |
| HighChol | 0 |
| CholCheck | 0 |
| BMI | 0 |
| Smoker | 0 |
| Stroke | 0 |





| | |
|----------------------|---|
| | 0 |
| HeartDiseaseorAttack | 0 |
| PhysActivity | 0 |
| Fruits | 0 |
| Veggies | 0 |
| HvyAlcoholConsump | 0 |
| AnyHealthcare | 0 |
| NoDocbcCost | 0 |
| GenHlth | 0 |
| MentHlth | 0 |
| PhysHlth | 0 |
| DiffWalk | 0 |
| Sex | 0 |
| Age | 0 |
| Education | 0 |
| Income | 0 |

dtype: int64
'Summary statistics for numerical columns:'

| | Diabetes_012 | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | ... | AnyHealthcare | |
|-------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------------|---------------|---------------|-----|---------------|---------------|
| count | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | ... | 253680.000000 | 253680.000000 |
| mean | 0.296921 | 0.429001 | 0.424121 | 0.962670 | 28.382364 | 0.443169 | 0.040571 | 0.094186 | 0.756544 | 0.634256 | ... | 0.951053 | |
| std | 0.698160 | 0.494934 | 0.494210 | 0.189571 | 6.608694 | 0.496761 | 0.197294 | 0.292087 | 0.429169 | 0.481639 | ... | 0.215759 | |
| min | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 12.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | ... | 0.000000 | |
| 25% | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 24.000000 | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 0.000000 | ... | 1.000000 | |
| 50% | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 27.000000 | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | ... | 1.000000 | |
| 75% | 0.000000 | 1.000000 | 1.000000 | 1.000000 | 31.000000 | 1.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | ... | 1.000000 | |
| max | 2.000000 | 1.000000 | 1.000000 | 1.000000 | 98.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | ... | 1.000000 | |

8 rows × 22 columns
'Distribution of Diabetes_012:'

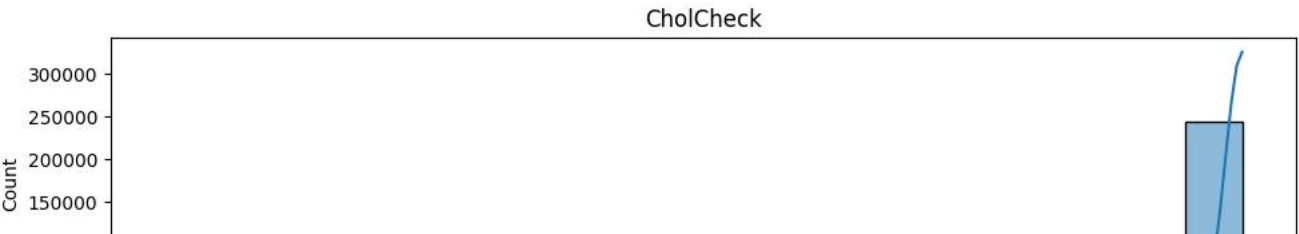
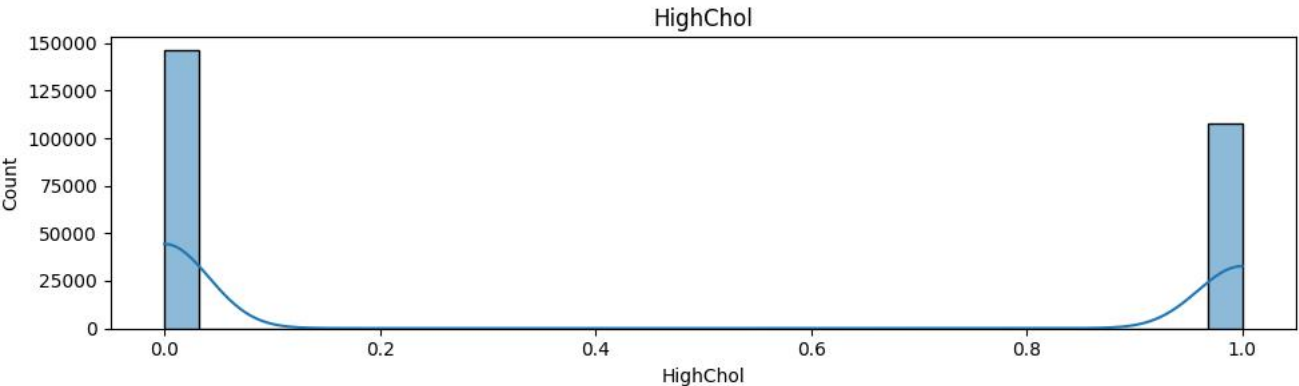
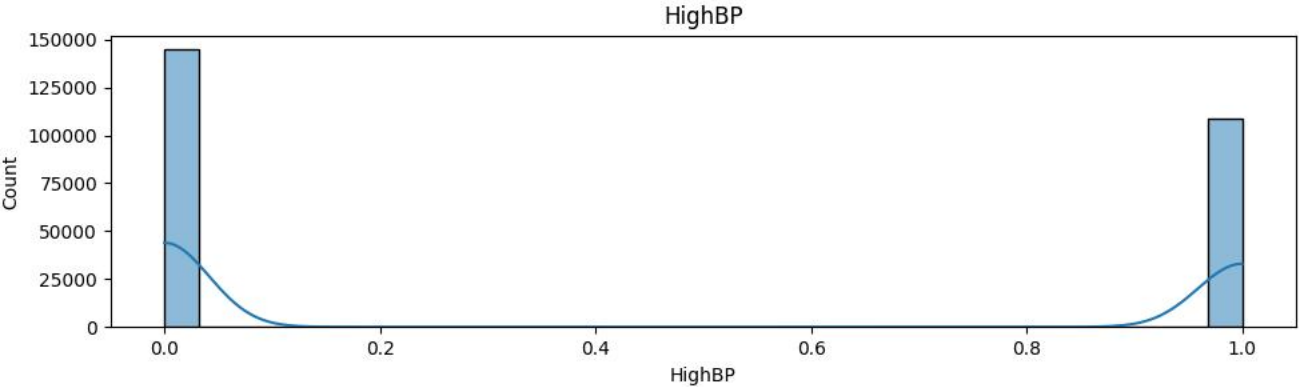
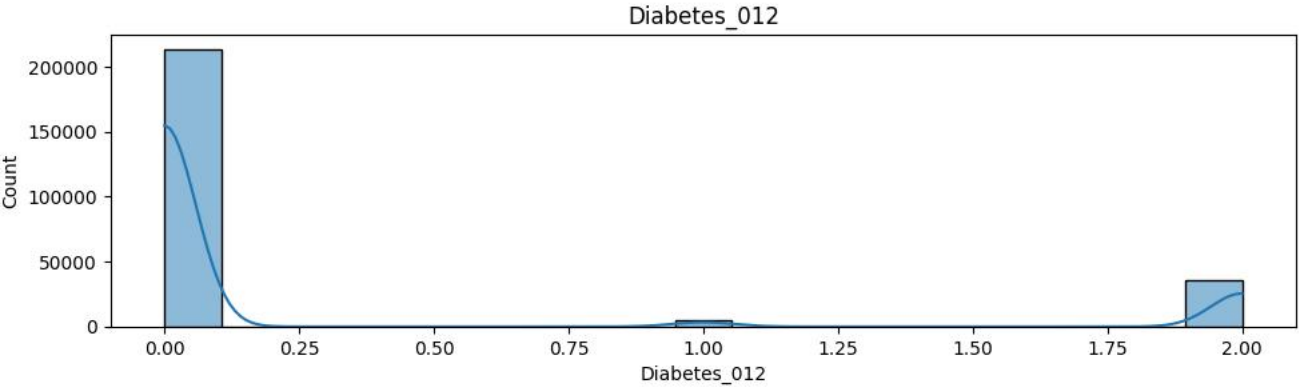
| Diabetes_012 | |
|--------------|--------|
| 0.0 | 213703 |
| 2.0 | 35346 |
| 1.0 | 4631 |

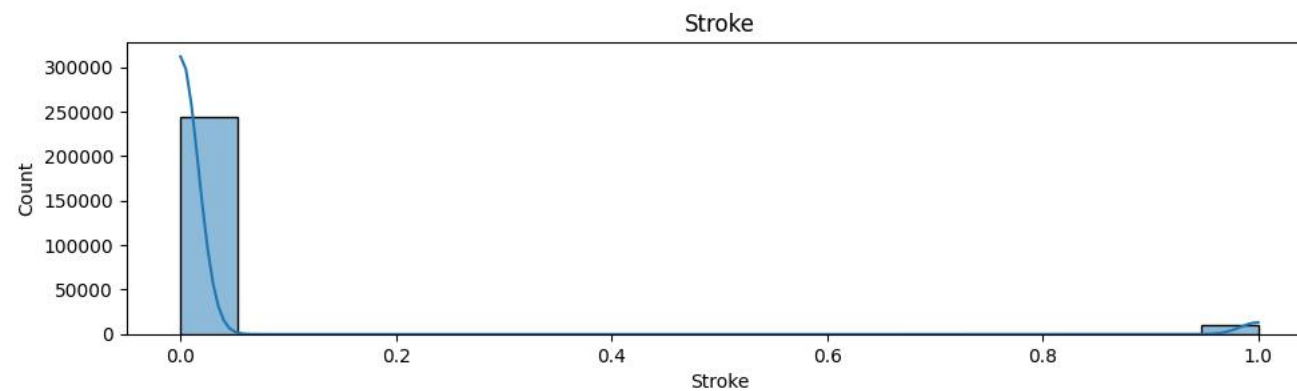
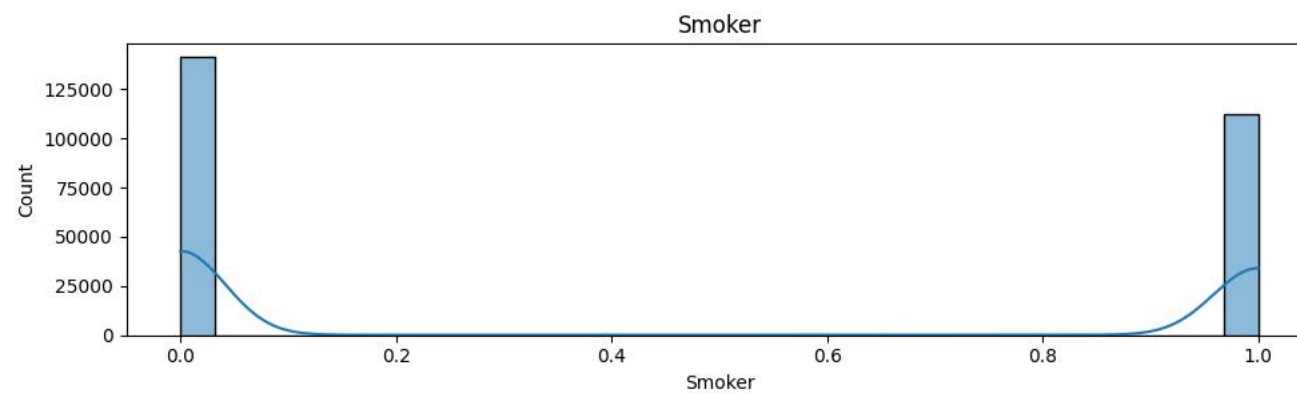
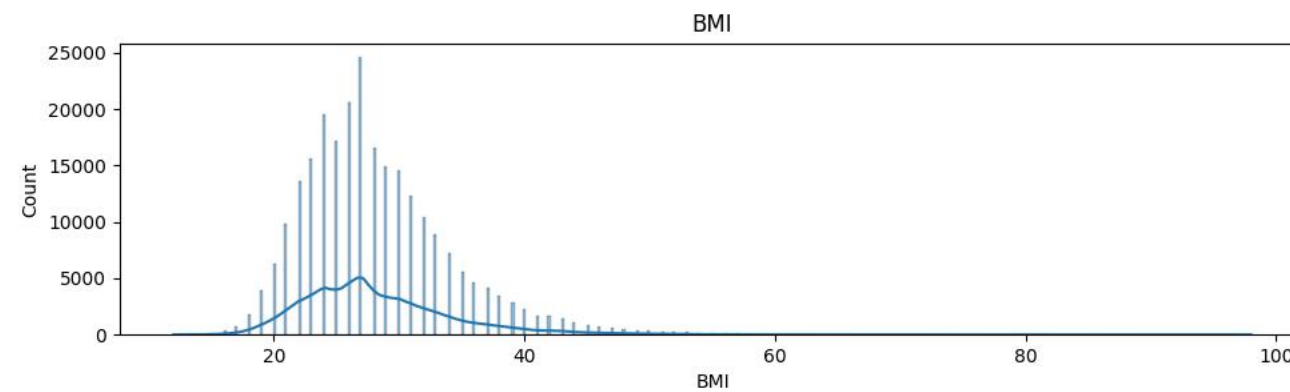
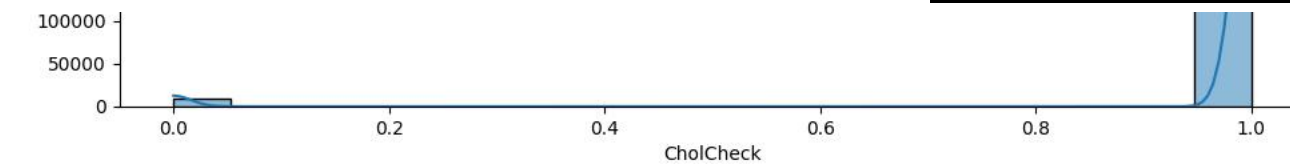




dtype: int64
'Histograms for numerical columns:'

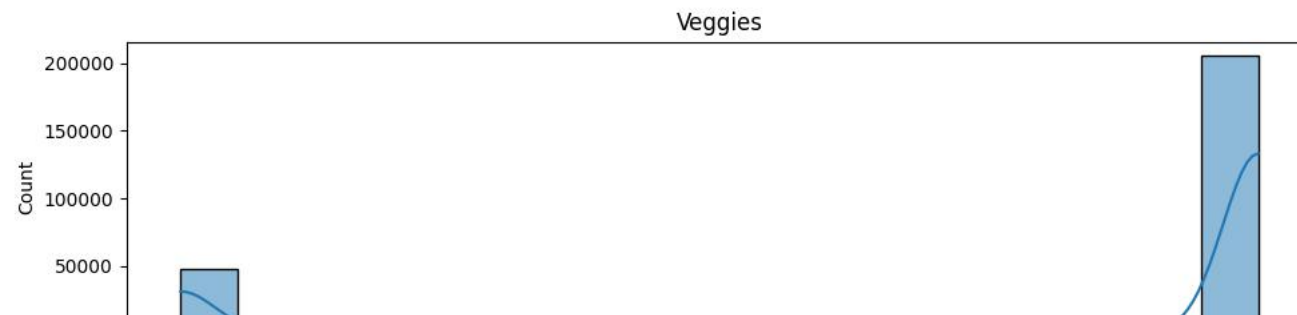
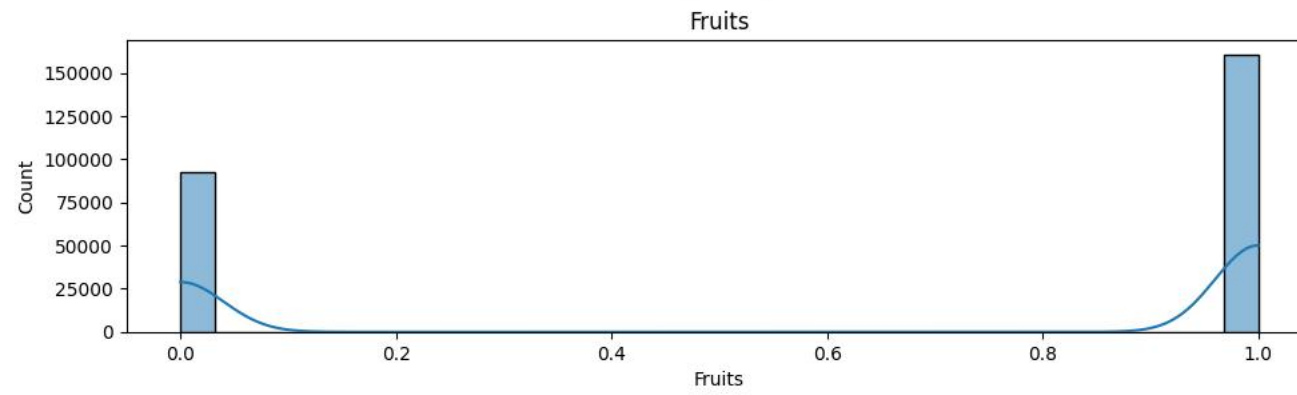
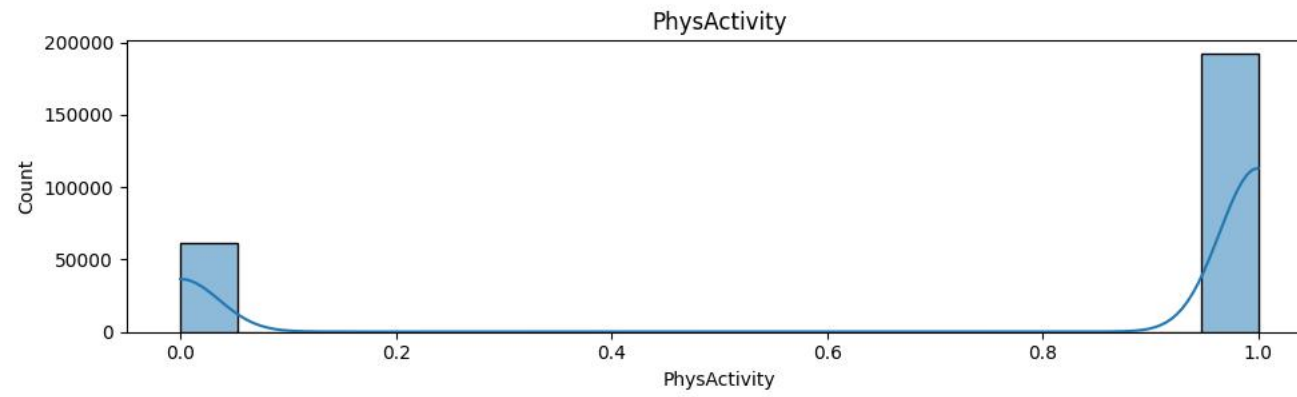
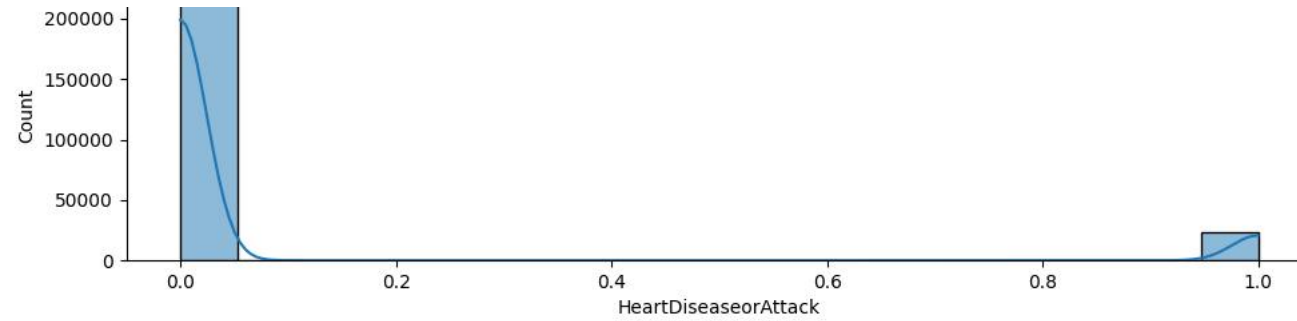


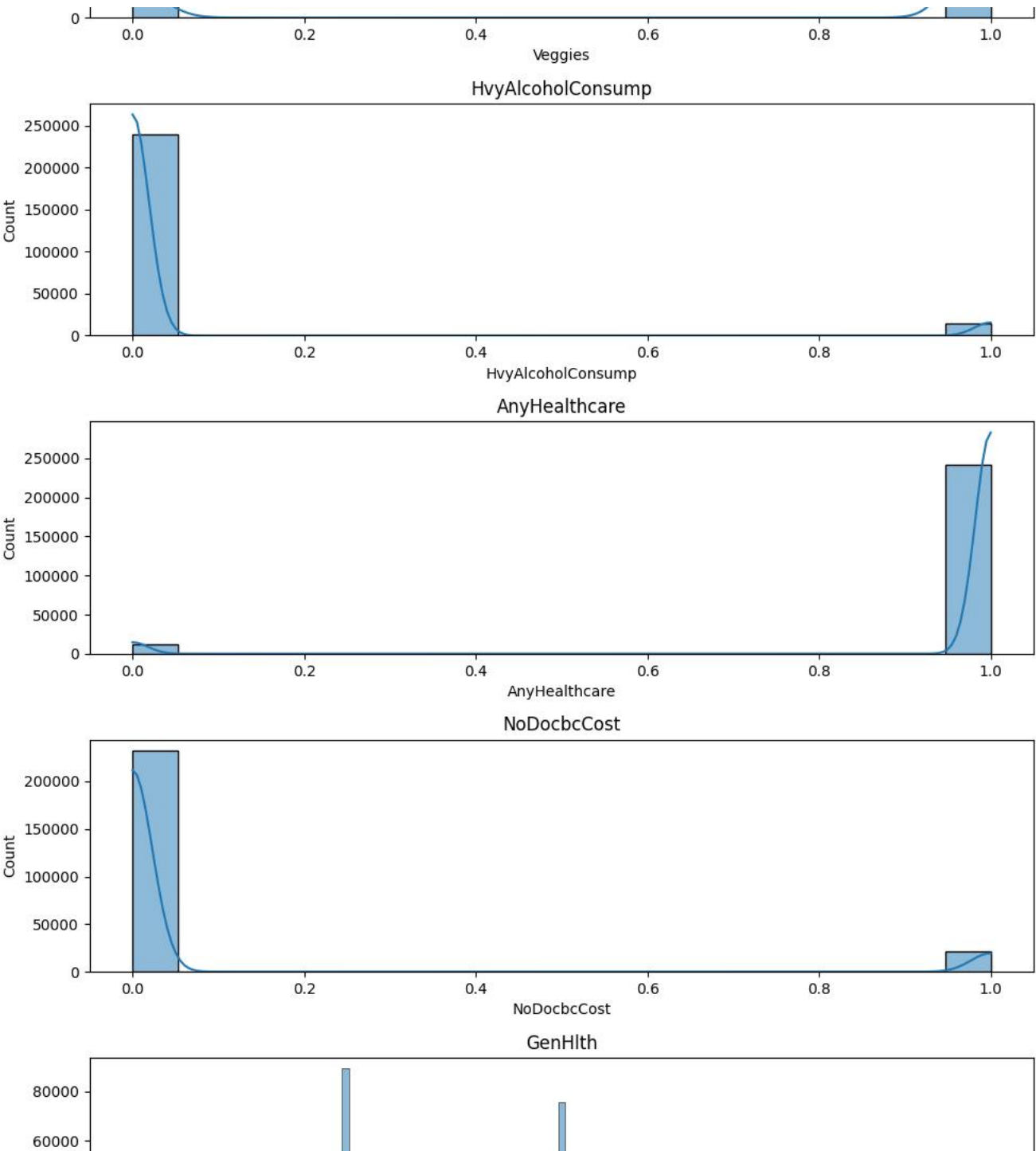


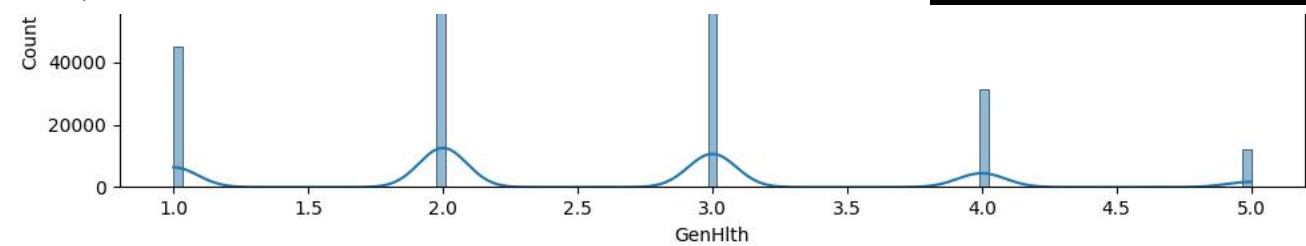


HeartDiseaseorAttack

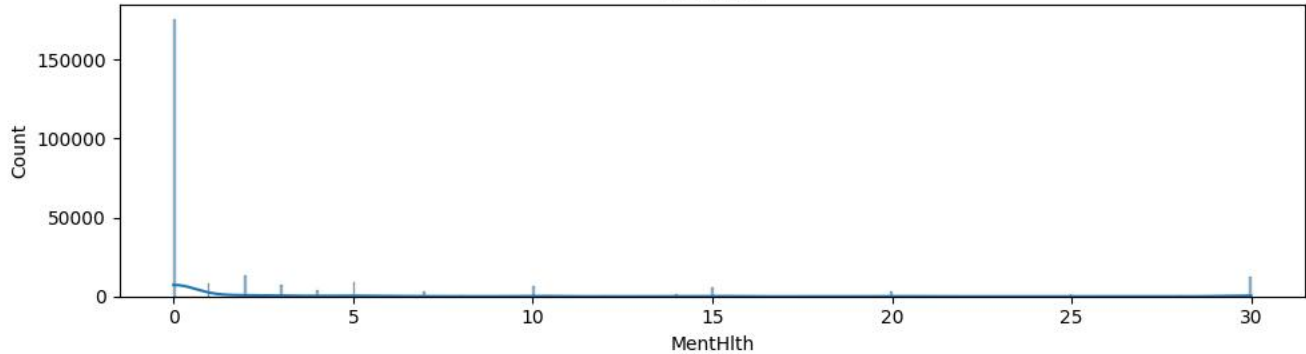




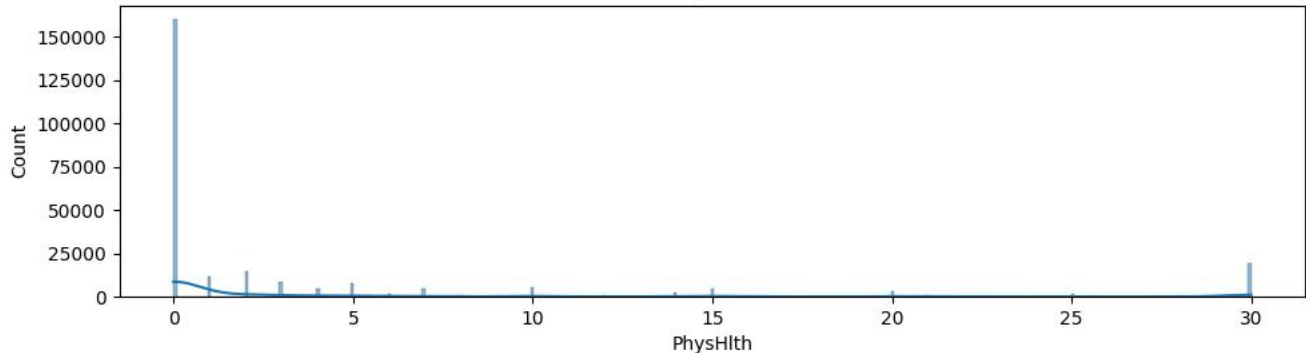




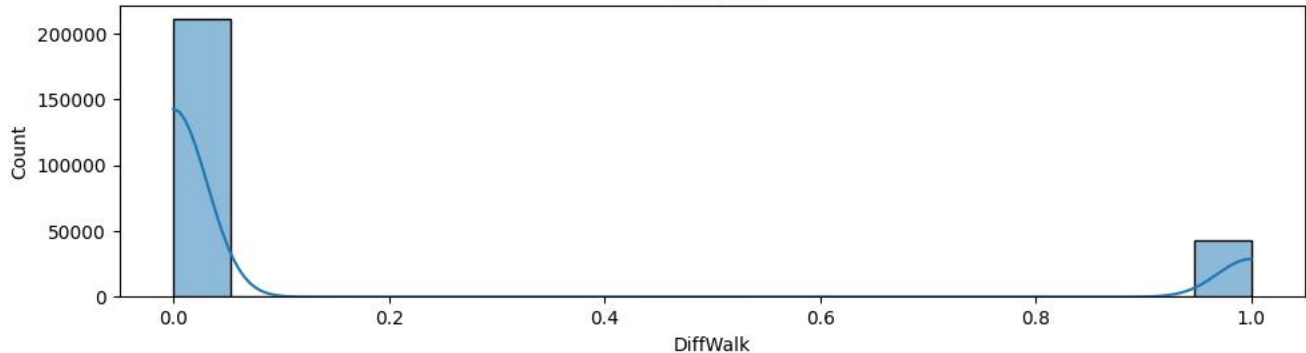
MentHlth

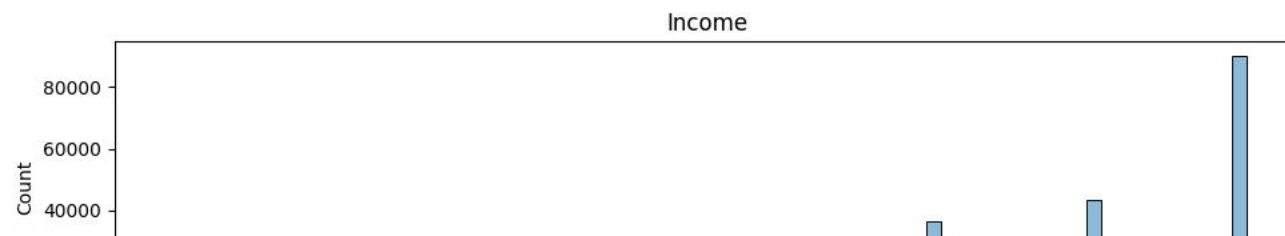
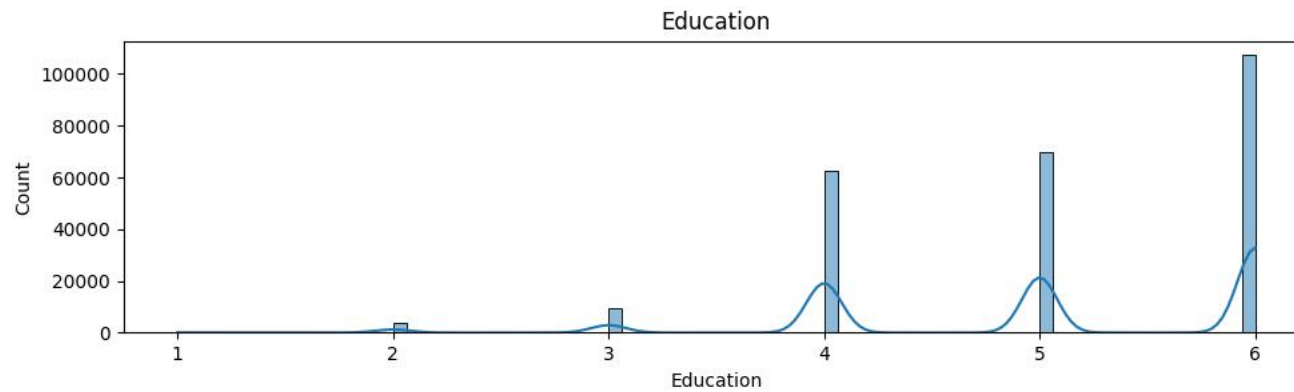
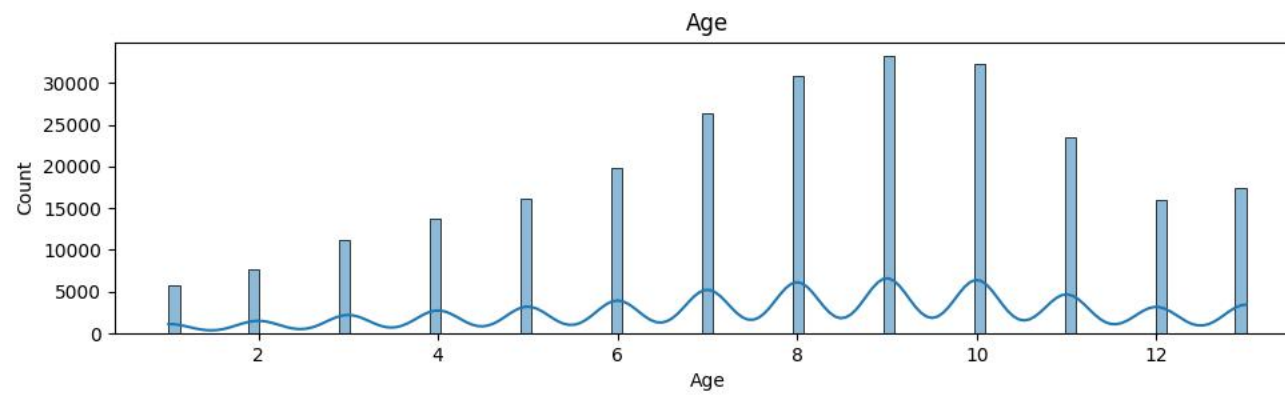
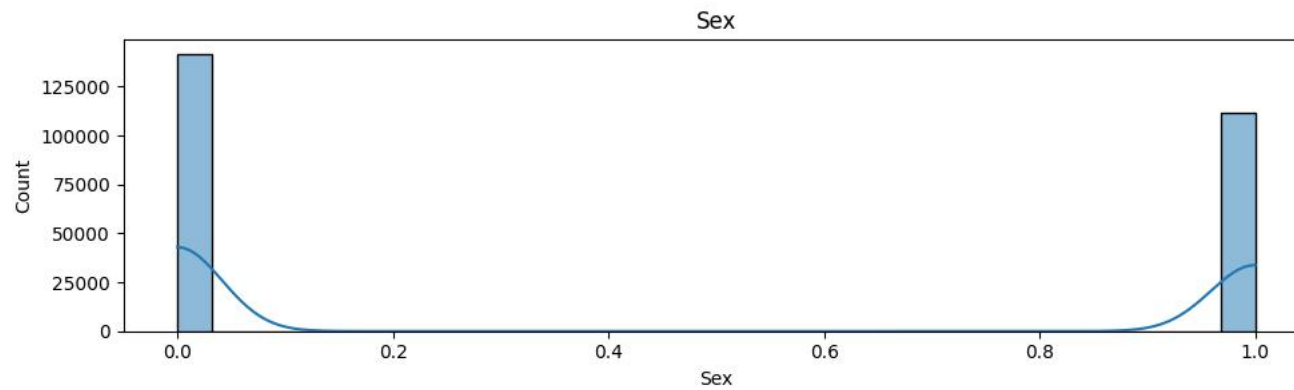


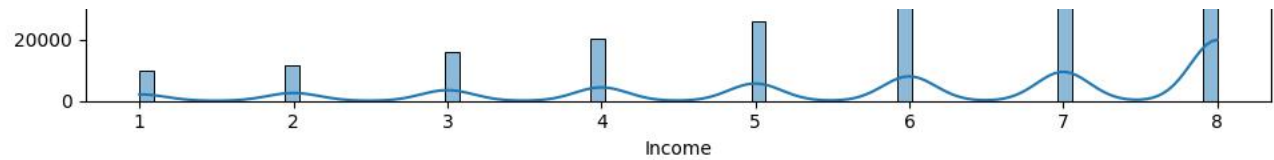
PhysHlth



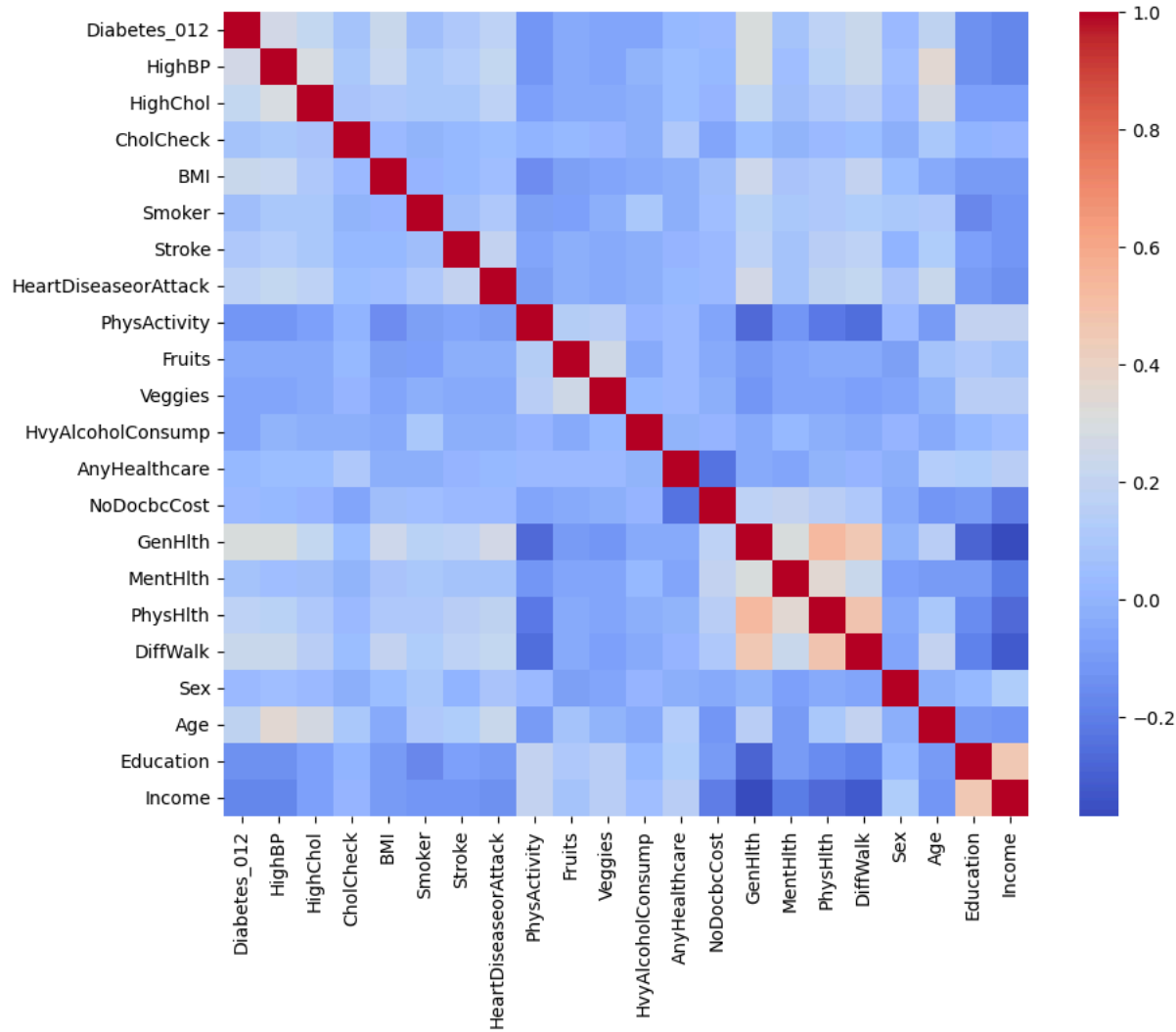
DiffWalk







'Correlation matrix:'



I tried to run the code, but got the following error: name 'np' is not defined

